The Architecture of Outer Admissibility

Unification, Maximality, and Residual Structure in Certificate Worlds

Nova Spivack

Independent Researcher

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Abstract

Outer certificates—compressed witnesses, abstraction layers, quotient representations, transport claims, upgrade witnesses—are routinely used as if they defined distinct admissibility worlds. This paper proves that, in a broad and disciplined certificate world, they do not: apparently different outer certificate forms collapse to one unified gate, force a derived higher layer above it, and leave stronger branch-specific content in a residual band. The unified outer gate is $G_{\mathrm{adm}} \equiv U_{\mathrm{pullback}}$. It implies the existence of an absolute frontier certificate, is maximal under the declared grammar of outer normal forms, and is stable against the nearest principled boundary candidate. Above the gate, a richer second-layer package is canonically forced without new hypotheses. Stronger branch-specific content sits above both levels as a residual band, partially reconstructible but irreducibly richer than the gate alone.

The result is a strict three-level architecture: *one outer gate of admission, one functorial second layer, one residual band.* Every boundary is characterized by a named, machine-checked theorem. All proofs are in Lean 4 with 0 `sorry`; exact theorem names and modules are given in the companion note and appendix.

Trust boundary: The three-level architecture is a theorem relative to the formalized certificate world and named Lean declarations; it is not an empirical claim about every deployed system without translation.

Keywords: outer admissibility, certificates, adequacy, Lean 4, layered architecture, formal verification

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Code/Data availability: `adequacy-architecture-lean`; `lake build`; `THEOREM_INVENTORY.md` and `MANIFEST.md` at repository root; companion `Adequacy_Theorem_Companion.tex` for concise theorem-name routing.

1 Premises, disagreement coordinates, and trust boundary

Where disagreement can (and cannot) live

Formal spine. Outer gate, second layer, and residual-band claims reference the machine-checked stack named in `THEOREM_INVENTORY.md`, not motivational prose alone.

Premise coordinates (for external readers).

- **Certificate-world grammar.** Universality is relative to the disciplined outer-normal-

form grammar formalized in Lean. *Logical effect:* Extending the grammar is a new mathematical task, not silent refutation of the original gate theorem.

- **Deployment bridge.** Mapping institutional "certificates" to formal predicates requires honest modeling; the paper does not certify informal APIs without that bridge. *Logical effect:* Skepticism about one application model leaves the core gate theorem intact.

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2 The problem

2.1 The phenomenon

Formal and applied systems routinely rely on outer certificates — external or interface-level predicates that claim to stand in for richer internal structure. These take many forms: *compressed witnesses* that claim to capture essential proof content, *abstraction layers* that claim to preserve enough structure for downstream reasoning, *quotient and interface representations* that project internal distinctions into a coarser type, *transport claims* that assert a structure moved along a reindexing retains its certificate content, and *upgrade witnesses* that present a certified object at one level as certifying a richer level.

In every case, the outer predicate makes the same claim: "the relevant content is here, at this interface, in this form." Without a theory of outer admissibility, such certificates are under-classified and often over-trusted: a pullback display, a branch-native pack, and an upgrade

witness appear to be three different kinds of outer certificate defining three different admissibility worlds.

The central discovery of this paper is that *they are not*. They are forced presentations of a single outer gate. Richer content does not spread sideways into alternative primitive modes — it is forced upward into a layered architecture above the gate. That is the architectural fact this paper proves.

The central problem is therefore not merely which outer certificates exist, but whether apparently different admissibility routes are genuinely different, or forced presentations of one underlying gate.

2.2 The adequacy burden

The Adequacy Architecture formalizes a prior and more fundamental constraint on this question. Before asking *which* outer certificates are honest, one must understand what it means for any account to be adequate.

The core insight is a theorem-level structural constraint:

The fundamental theorem (informal)

If a system is rich enough, adequacy has a cost. You cannot freely flatten, summarize, certify, or route around the burden of adequacy without that burden reappearing somewhere else.

Four machine-checked consequences form the foundation for the outer admissibility results:

1. **No free adequate collapse.** If a representation is adequate in a non-trivial mode, it cannot erase all burden without paying elsewhere [1].
2. **Burden relocates; it does not vanish.** Honest simplification moves structural obligations to another layer or carrier. The obligations are not deleted; they are relocated.
3. **Route architecture is forced.** Once adequacy and burden are genuinely entangled, route and finality claims become part of what must be exhibited. They cannot be treated as decorative.
4. **Every forgetful morphism has a residual story.** Any map that loses information has a canonical residual structure recording what was lost, where it went, and what still observes it. This is the machine-checked form of the diagnostic principle: if the burden is invisible, suspect the map.

These four facts are not background. They are the structural reason that outer certificate worlds cannot be flat: adequacy forces burden, burden forces route architecture, and maps that compress structure have residuals that cannot be erased. The outer gate theorem is the culmination of this chain for the specific world of typed outer certificate predicates.

2.3 The classification problem

Given the prior foundation, the outer classification problem becomes precise.

An outer certificate predicate $C : \mathsf{Type} \to \mathsf{Prop}$ evaluated at a type-indexed slot γ claims: "this slot is genuinely admissible — it connects to the frontier of the adequacy landscape." There

are many ways such a claim could be made: via pullback displays, via full branch-specific packs, via upgrade witnesses, via grammar-generated composites.

The natural question is whether these are genuinely different admissibility conditions, or presentations of the same one. But that question leads immediately to three more:

(1) **Do admissibility routes unify?** Are different presentations of admission actually the same gate?

(2) **What is forced above the gate?** Does admission unlock richer structure canonically, or only with additional hypotheses?

(3) **What is irreducibly residual?** Is there branch-specific content that is stronger than the gate but cannot be absorbed into it or into the second layer?

This paper answers all three questions with a single architectural fact:

Apparently different outer certificate forms do not define different admissibility worlds. Under disciplined adequacy constraints, they collapse to one outer gate, force a derived higher layer above it, and leave stronger branch-specific content in a residual band that cannot be absorbed into either level.

3 Why this is hard and important

3.1 The naïve answer fails

A naïve answer to the outer classification problem would be: any predicate that happens to hold at some γ counts as admissible. That answer is empty — it makes the notion of admissibility meaningless.

A better but still naïve answer would be: enumerate the known certificate constructors and call each of them admissible. But this *pluralist* answer raises its own problem: it produces a collection of admissibility conditions without answering whether they form one class or many, what their relationship is, and where the ceiling is.

3.2 What makes the problem non-trivial

The problem is difficult in both directions. From above, the admissibility family is presented by structurally different constructor arms. From below, the unified witness U_{pullback} is not vacuous: it requires a genuine lawful pullback along a compare map, realized in the library by distinct nontrivial proof routes [1]. The collapse theorem therefore compresses real structural diversity into a gate that itself carries genuine content.

3.3 What the architecture really shows

The deeper import is not classificatory but architectural. The paper proves that outer certificate worlds are forced into three bands: one admissibility ceiling within the declared grammar, one derived layer above admission, and one residual band that stronger branch-specific content cannot escape.

Three consequences make this concrete.

The outer gate is the unique admissibility ceiling. Once the normal-form grammar is fixed, no class generated from it can exceed U_{pullback}: the gate is not the highest thing found, it is the highest thing there is within the grammar (§6).

The second layer cannot be reached from below. No admissibility route produces the second-layer content directly; it is produced canonically *after* admission, with no new hypotheses — what the gate forces into existence above itself (§7).

The residual content cannot be absorbed. The branch-specific residual condition carries a genuine irreducible piece that the outer gate alone cannot supply, blocking its absorption into either level (§8).

When someone presents an outer predicate as an admissibility condition, the architecture asks four questions whose answers determine its position in the forced three-level architecture: does it imply an absolute frontier certificate (gate consequence); is it grammar-generated (ceiling test); does it unlock a second layer or is it second-layer content (layering test); is it primitive or residual (residuality test)?

4 The central discovery

4.1 What the theorems say

Before the formal machinery, the central picture can be stated plainly.

Central discovery

Outer certificate worlds are not merely plural collections of admissibility tests. They are **forced into a layered architecture** with one unified gate, derived higher structure, and residual stronger content. Specifically:

(1) There is a **single unified outer gate of honest admission**. Multiple finite admissibility presentations — pullback display, full branch pack, upgrade witness — collapse to one gate $G_{\text{adm}} \equiv U_{\text{pullback}}$.

(2) The unified gate **has a real consequence**: it implies the existence of an absolute frontier certificate.

(3) The gate is **maximal under the declared grammar** (NF0–NF3): no grammar-generated class exceeds it, and the nearest non-generated boundary candidate does not force expansion.

(4) Above the gate, a **functorial second layer** is canonically produced from the same witness, without additional hypotheses. It is not a new admissibility mode; it is what admission *unlocks*.

(5) **Stronger branch-specific content is genuinely residual**: it does not collapse into Level 1 or Level 2, but carries its own irreducible obstruction.

4.2 Three forced levels, not three choices

The five points above are not a tidy taxonomy. They are a structural impossibility result. The three levels are not choices. They are three different kinds of structural content, each forced into

its own band by the logic of admissibility: the grammar-cut maximality theorem blocks escape from Level 1 by composition, the layering theorem blocks direct access to Level 2 from below, and the residuality result blocks absorption of Level 3 into either level above.

The solved architecture has exactly three levels:

$$\textbf{Level 1: } G_{\mathrm{adm}} \equiv U_{\mathrm{pullback}} \qquad \textbf{Level 2: } U_{\mathrm{pullback}} \Rightarrow L_2 \qquad \textbf{Residual: } \mathrm{StageD}$$

Level 1 is admission, Level 2 is a canonically derived higher package, and the residual band contains stronger branch-specific content that neither collapses to the gate nor belongs to the functorial second layer.

Every boundary is theoremically sharp. The three levels are not a conceptual overlay on the results; they are exactly what the results force. The gate, the second layer, and the residual band are separated by named machine-checked implications and non-implications.

5 The formal world and disciplined scope

5.1 The type-indexed outer world

Fix a universe of types. An *outer certificate predicate* is a predicate $C : \mathsf{Type} \to \mathsf{Prop}$. We evaluate such predicates at a type-indexed slot γ.

Definition 5.1 (Finite admissibility gate — G_{adm})**.** $G_{\mathrm{adm}}\,\gamma$ (`Program1FiniteGAdm`) holds when γ is witnessed by one of three constructor arms:

1. A pullback display arm: `Program1AdmissibilityPullbackDisplayWitness` γ.
2. A branch-native full pack arm (connecting the infinity-compression architecture to the standard carrier): `Program1IcStageDFullPackWitness` γ.
3. A representation-calculus upgrade witness arm: `Program1CertifiedUpgradeWitness` γ.

Definition 5.2 (Pullback display witness — U_{pullback})**.** $U_{\mathrm{pullback}}\,\gamma$ (`Program1AdmissibilityPullbackDisplayWitness`) holds when there exists a certified frontier representation obtained by pulling back a lawful adequacy architecture along a compare map at γ.

U_{pullback} is not vacuous. The branch-native arm requires a non-trivial compression carrier pullback. The carrier-swap arm requires an involutive compare-map cleaving argument. Both are proved as separate machine-checked theorems [1].

5.2 The outer normal-form grammar

The outer grammar is the closure of the following four normal forms under declared grammar rules:

NF0. Bare pullback display.

NF1. Certified frontier row.

NF2. Upgrade witness.

NF3. Composition of the above.

$G_{\text{gen}}\, C$ means $C : \mathsf{Type} \to \mathsf{Prop}$ is generated by this grammar. NF4 (S_1^{lawful}), the lawful S1 frontier row predicate, lies outside the grammar. The maximality theorem (§6) shows that nothing grammar-generated exceeds the outer gate, and the NF4 boundary analysis shows that even this nearest non-generated candidate does not force expansion.

The point of the grammar is not convenience. It is the smallest disciplined outer language in which one can ask a meaningful maximality question without collapsing into unconstrained predicate pluralism.

5.3 Notation

Symbol	Lean name	Meaning
$G_{\text{adm}}\, \gamma$	`Program1FiniteGAdm`	finite admissibility gate
$U_{\text{pullback}}\, \gamma$	`Program1AdmissibilityPullbackDisplayWitness`	pullback display witness
$G_{\text{gen}}\, C$	`Program1GrammarGeneratedOuterGate`	grammar-generated
$\text{AbsRawS1}(P, B)$	`AbsoluteFrontierRawS1`	absolute frontier S1 certificate
$S_1^{\text{lawful}}\, \gamma$	`Nonempty (S1LawfulFrontierRow γ)`	NF4 win predicate
$L_2\, \gamma$	`pullbackDisplay_with_host_summitFE3`	second-layer package
$\text{StageD}\, \gamma$	`icSpec031_stageD_resolution_pack`	branch-specific residual pack

5.4 Foundational dependencies and scope

The outer gate program builds on two prior machine-checked results. First, the highest reachable summit H_{final} is packaged as a structured object by `highestReachable_summit_at` (*Lawful/FinalConditionalSummit.lean*); the unification theorem shows all three constructor arms of G_{adm} supply this target through U_{pullback}. Second, the residual-geometry library provides the general principle that every forgetful morphism has a fundamental residual structure — the machine-checked form of "if the burden is invisible, suspect the map" — of which the residual band (§8) is a specific instance.

Scope:

- No claim of unconstrained $\forall P\, B$. An unconditional frontier result for arbitrary predicate pairs is the next genuine research frontier; it is not part of the present theorems (see §12 and [1]).
- Maximality is grammar-cut: within the declared normal-form grammar (NF0–NF3), not over all possible predicates.
- The NF4 boundary analysis is a boundary report, not a completeness claim.
- The residual band (§8) is branch-specific. Its irreducibility is an honest finding, not a deficiency.
- All theorems below are machine-checked; 0 `sorry`. Toolchain: Lean 4 v4.29.0-rc6, Mathlib v4.29.0-rc6 [3].

6 Collapse to the outer gate

6.1 The unification theorem

The first major result shows that the apparent plurality of admissibility routes collapses.

Theorem 6.1 (Unification)**.** *For all* γ, $\quad G_{\text{adm}}\, \gamma \iff U_{\text{pullback}}\, \gamma$.

```
program1FiniteGAdm_iff_program1AdmissibilityPullbackDisplayWitness :
  forall gamma, G_adm gamma <-> U_pullback gamma
-- Module: Instances/Program1MetaUnificationMU1.lean
```

This is the first summit of the paper: presentations that look like different outer worlds turn out to be only different routes into one and the same admission gate.

This is nontrivial in both directions.

Three arms collapse (the non-trivial direction). Each of the three constructor arms of G_{adm} must be shown to produce a U_{pullback} witness:

- The pullback arm is immediate (first constructor).
- The branch-native full pack arm threads through a non-trivial compression carrier pullback (the infinity-compression compare-map representation, proved separately in the Lean library).
- The upgrade witness arm factors through the certified summit-row representation via the representation-calculus gate.

The gate is not empty (the honesty direction). U_{pullback} requires a genuine lawful adequacy architecture pulled back along a compare map at γ. Four concrete witnesses have been constructed — using the infinity-compression pullback, an identity-level corpus instantiation, a carrier-swap corpus instantiation, and a parity quotient on a three-element type — each requiring distinct non-trivial arguments [1].

6.2 The absolute frontier consequence

Theorem 6.2 (Absolute frontier S1 from admission)**.** *For all γ,*

$$G_{\mathrm{adm}}\,\gamma \;\Rightarrow\; \exists P\,B,\ \mathrm{AbsRawS1}(P,B).$$

```
program1FiniteGAdm_implies_exists_absoluteFrontierRawS1 :
  forall gamma, G_adm gamma -> exists P B, AbsoluteFrontierRawS1 P B
-- Module: Instances/Program1MetaUnificationMU1.lean
```

This is the decisive outer payoff: admission is not merely a classificatory status. It forces the existence of an absolute frontier certificate.

Admission at any γ witnesses the non-emptiness of the absolute frontier. The absolute frontier $\mathrm{AbsRawS1}(P,B)$ is the frontier of the adequacy landscape: it says the highest reachable summit exists for some concrete predicate–burden pair, as formalized in the machine-checked absolute-frontier program [1].

7 Maximality and the grammar boundary

7.1 Grammar-cut maximality

Theorem 7.1 (Grammar-cut maximality)**.** *Let $C : \mathsf{Type} \to \mathsf{Prop}$. If $G_{\mathrm{gen}}\,C$, then $\forall\,\gamma,\ C\,\gamma \Rightarrow U_{\mathrm{pullback}}\,\gamma$.*

```
program1_outer_grammar_cut_maxA :
  forall (C : Type -> Prop),
    Program1GrammarGeneratedOuterGate C ->
    forall gamma, C gamma -> U_pullback gamma
-- Module: Instances/Program1OuterGateGrammar.lean
```

Theorem 6.1 establishes that U_{pullback} is the *ceiling* of the declared grammar. Any outer predicate generated by composing, lifting, or grammar-producing from NF0–NF3 stays within U_{pullback}.

Without this theorem, the unified gate would be a lower bound: something that everything we know satisfies, but which could in principle be exceeded by some more expressive grammar-generated class we have not tried. With it, the gate becomes a ceiling rather than a lower bound. Within the declared grammar, there is no higher admissible form. This turns the outer gate from a successful unification into a true maximality theorem.

7.2 The NF4 boundary finding

The nearest principled candidate for a broader outer class is the NF4 predicate $S_1^{\text{lawful}}\,\gamma$, which asserts the non-emptiness of a lawful S1 frontier row at γ.

Theorem 7.2 (NF4 outer soundness)**.** *For all* γ, $\quad S_1^{\text{lawful}}\,\gamma \Rightarrow$ *`program1OuterLayerBLaw`*γ.

```
Program1OuterCertificateSound_program1OuterWinS1LawfulNonempty :
  forall gamma, S1Lawful gamma -> program1OuterLayerBLaw gamma
-- Module: Instances/Program1MaximalityMA1.lean
```

The NF4 boundary analysis records three facts jointly:

1. S_1^{lawful} is outer-sound: it implies the outer layer law (Theorem 6.2).
2. S_1^{lawful} is *not* grammar-generated (not NF0–NF3).
3. There is *no proved universal* $S_1^{\text{lawful}}\,\gamma \Rightarrow U_{\text{pullback}}\,\gamma$.

NF4 therefore remains a logical pocket: sound where instantiated, but not promoted to the operative outer gate.

7.3 Counterexample: S1-shape is necessary

Before stating what the NF4 boundary means, we note that an explicit machine-checked counterexample shows the S1-shape condition is not vacuous:

```
not_s1LawfulFrontierRow_counterexample_pair :
  forall ..., Not (Nonempty (S1LawfulFrontierRow counterexamplePair))
-- Module: Portability/CertifiedFrontierRow.lean (S1ShapeNecessityCounterexample)
```

There exist explicit pairs (P, B) where no lawful S1 frontier row exists. This means the S1-shape requirement in the architecture is load-bearing: it is not trivially satisfied. The grammar boundary separates something real from something vacuous.

7.4 What the NF4 boundary means

The NF4 finding matters in two directions simultaneously.

Downward: it confirms the architecture is not trivially all-encompassing. The grammar boundary is real. There exist predicates (like S_1^{lawful}) that are not grammar-generated and not universally equal to U_{pullback}.

Upward: NF4 did not force expansion. The nearest candidate for "maybe there is a broader class" turned out not to yield one. The architecture is stable at its current boundary — not because we could not find anything stronger, but because the nearest candidate was examined and did not fit.

Taken together, grammar-cut maximality and NF4-C show that U_{pullback} is neither an accidental lower bound nor a vacuous universal. It is the correct operative outer gate at the present level of disciplined structure.

8 The second layer above admission

8.1 The second-layer consequence

One of the most structurally important results is not about the outer gate itself. It is about what the outer gate unlocks.

Theorem 8.1 (Functorial second layer)**.** *For all γ, $U_{\text{pullback}}\,\gamma \;\Rightarrow\; L_2\,\gamma$ (and therefore $G_{\text{adm}}\,\gamma \;\Rightarrow\; L_2\,\gamma$).*

```
program1_layer2PullbackDisplay_with_host_summitFE3_of_U_pullback :
  forall gamma, U_pullback gamma -> L2 gamma
program1_layer2PullbackDisplay_with_host_summitFE3_of_program1FiniteGAdm :
  forall gamma, G_adm gamma -> L2 gamma
-- Module: Instances/Program1LayeredStructureLS1.lean
```

The L_2 package (`pullbackDisplay_with_host_summitFE3`) combines the pullback display with a host-side summit certificate. It is obtained by projecting from the same witness that gives $U_{\text{pullback}}\,\gamma$ — no additional hypotheses are required.

8.2 Why this is structurally important

L_2 is not a new constructor of G_{adm}, not grammar-generated, and does not imply G_{adm} — it is downstream. It is canonically produced from any U_{pullback} witness without choice or additional hypotheses.

This means: an outer certificate is not just a stamp of approval. It is an entry ticket to a richer world. Admission is not a dead end; it is the beginning of a richer theory. The second-layer package combines the pullback display with a host-side summit certificate, giving a richer simultaneous view than admission alone provides. The outer certificate world has strict vertical structure: Level 1 (admission) below, Level 2 (its canonical functorial consequence) above. These are not the same level, and nothing in Level 2 can be accessed without first passing through Level 1. This is the paper's second major architectural discovery: admission does not terminate the theory; it opens a canonically richer one. The significance is architectural: admission does not terminate the theory. It canonically generates a richer one.

9 Stage D and the residual band

9.1 What the residual condition is

The residual condition StageD is a branch-specific pack of requirements connecting the infinity-compression architecture to the highest reachable summit at the standard carrier. It decomposes into four conjuncts:

Theorem 9.1 (Residual condition decomposition)**.** *For all γ,*

$$\mathrm{StageD}\,\gamma \iff F_1(\gamma) \wedge F_2(\gamma) \wedge F_3(\gamma) \wedge F_4(\gamma)$$

where F_1 = obstruction entry, F_2 = absolute-frontier S1 conjunct, F_3 = bypass, F_4 = cascade.

Each of the four conjuncts has a named machine-checked lemma (see the companion note for exact names). Facets F_1, F_3, F_4 are reconstructible from named hypotheses. F_2 is the irreducible residue.

9.2 The residuality result

Proposition 9.2 (Residual condition residuality)**.** *(a) $U_{\mathrm{pullback}}\,\gamma$ does not imply* StageD γ *in general.*

(b) StageD γ *does not imply $U_{\mathrm{pullback}}\,\gamma$ without the F_1 and bypass arms.*

(c) F_1, F_3, F_4 are reconstructible from named hypotheses.

(d) F_2 (the absolute-frontier S1 conjunct) is the irreducible residue: it does not follow from $U_{\mathrm{pullback}}\,\gamma$ alone.

9.3 Why residuality matters

The residual band is one of the paper's key findings. It identifies the exact point at which stronger branch-specific content ceases to be governed by the outer gate or its functorial second layer. Proposition 8.2 closes off the tempting reclassification: StageD is not primitive to the outer gate. It is not part of the second layer. It is branch-specific extra structure with its own internal organization (the four conjuncts) and its own irreducible obstruction (F_2). The genuinely stronger branch-specific conditions do not sneak back into Level 1 or Level 2 — they live in their own residual band, partially accessible via named lemmas, with an irreducible core.

The residual band is therefore not a defect in the architecture but one of its defining boundaries. It marks exactly what the unified outer gate does not control.

The general principle underlying this finding is machine-checked throughout the architecture: every forgetful morphism has a canonical fundamental residual structure recording what was lost and what still observes it. F_2 is the specific instance of this principle for the infinity-compression branch: the map from the outer gate to the compression architecture forgets exactly the F_2 content.

10 The complete architecture

10.1 The three-level picture

```
NF4 (S1Lawful)  -- outer-sound; not grammar-generated;
                   no universal S1Lawful -> U_pullback
      ^
-------------------------------------------------------
Residual: branch-specific condition  (partial basis;
                     F2 is irreducible w.r.t. U_pullback)
      ^
-------------------------------------------------------
Level 2: L2 = pullbackDisplay_with_host_summitFE3
               [functorial from U_pullback; not a new outer mode]
      ^
-------------------------------------------------------
Level 1: G_adm <=> U_pullback  (outer gate; maximal NF0--NF3)
      v
exists P B, AbsoluteFrontierRawS1 P B
```

Every arrow is a named machine-checked theorem. The horizontal rules are logical boundaries characterized by what is and is not derivable from U_{pullback} alone.

10.2 The integrated theorem

Main theorem of the outer admissibility program

For the disciplined outer admissibility world:

(I) Collapse. All finite admissibility presentations — the pullback display arm, the branch-native full pack arm, the upgrade witness arm — are logically equivalent: $G_{\text{adm}} \equiv U_{\text{pullback}}$ (Theorem 5.1). There is one outer gate.

(II) Consequence. The outer gate has a real payoff: $G_{\text{adm}}\,\gamma \Rightarrow \exists P\,B,\ \text{AbsRawS1}(P, B)$ (Theorem 5.2). Admission implies an absolute frontier certificate exists.

(III) Maximality. The outer gate is the ceiling of the declared grammar: no predicate generated from NF0–NF3 exceeds U_{pullback} (Theorem 6.1). The nearest boundary candidate (NF4) does not force expansion: the grammar boundary is real and stable (§6).

(IV) Forced second layer. A richer second-layer package L_2 is canonically produced from any admission witness, with no additional hypotheses (Theorem 7.1). The second layer is not a new admissibility mode; it is what admission forces into existence above itself.

(V) Residual band. Stronger branch-specific content StageD is not primitive to the outer gate, is not part of the second layer, and carries a genuine irreducible obstruction (Proposition 8.2). It is classified as residual, not dismissed.

Points I–III identify the unique outer gate and its ceiling. Point IV shows that admission canonically generates a richer second layer rather than terminating at classification. Point V

shows that stronger branch-specific structure remains real but residual. Together they yield the full forced architecture of outer admissibility: one gate, one derived layer, one residual band.

> *Apparently different outer certificate forms do not define different admissibility worlds. Under disciplined adequacy constraints, they collapse to one outer gate, force a derived higher layer above it, and leave stronger branch-specific content in a residual band that cannot be absorbed into either level.*

The architecture could not have come out flatter. Every attempt to flatten it—to conflate the levels, to absorb the residual, to escape the gate by grammar composition—is blocked by a proved theorem. The three levels are the minimum that the logic of admissibility forces into existence.

10.3 What every arrow earns

The diagram above is not a schematic. Three of the arrows carry the full weight of the architecture:

- $G_{\text{adm}} \iff U_{\text{pullback}}$:
  `program1FiniteGAdm_iff_program1AdmissibilityPullbackDisplayWitness`
- $G_{\text{gen}} \subseteq U_{\text{pullback}}$:
  `program1_outer_grammar_cut_maxA`
- $U_{\text{pullback}} \Rightarrow L_2$:
  `program1_layer2PullbackDisplay_with_host_summitFE3_of_U_pullback`

The full Lean name list for all arrows is in Appendix A.

11 Robustness and breadth of applicability

11.1 Why the gate is robust

Four structurally heterogeneous admissibility presentations have already been shown to collapse to U_{pullback}: the infinity-compression pullback arm (compression carrier along a constant compare map), two carrier-level corpus instantiations (identity and carrier-swap, the latter via involutive cleaving), a parity quotient on a three-element type, and the certified upgrade witness arm.

Beyond these instances, grammar-cut maximality says the result holds for *any* predicate generated by the outer normal-form grammar — not just the specific ones proved above, but the entire closure under NF0–NF3. The gate is therefore not the highest specific case found; it is the highest provable ceiling within the declared grammar.

11.2 Honest boundaries

The architecture does not claim to govern all outer predicates. NF4 exists, is sound, and is not grammar-generated — proving the grammar boundary is real. The NF4 finding then provides an independent robustness check: the nearest candidate for a broader class was examined and did not force expansion. The grammar boundary is genuine, not accidental.

Equally, the residual condition is not claimed to be unreachable. It has a partial basis and an irreducible residue; it is classified precisely, not dismissed. The architecture is therefore broad

in a structural, not merely enumerative, sense. It governs a disciplined world of outer certificate forms and proves that this world organizes itself into one gate, one derived layer, and one residual band.

12 Breadth: what this architecture governs

12.1 What class of certificate worlds is governed

The architecture governs any certificate world where:

- There is a notion of adequacy with real, non-erasable burden.
- There is a lawful class of architectures supporting stratified adequacy, bottleneck structure, and bridge relationships between layers.
- There are outer certificate predicates evaluated at type-indexed slots.
- There is a grammar of outer normal forms.

Within such a world, the entire theorem stack applies: the outer gate unifies, the second layer is forced above it, and residual content is classified and located. In this sense the paper is not merely classifying a set of examples. It is identifying the architecture of admissibility for a broad disciplined world of outer certificate forms.

More specifically, the architecture governs outer certificate forms built from any combination of:

- Pullback-style representability along a compare map.
- Finite admissibility presentations with multiple constructor arms.
- Grammar-generated admissibility schemes under declared normal forms.
- Quotient and interface certificate forms.
- Upgrade or witness-promotion routes.
- Functorial layered consequence packages.

This is not a narrow or idiosyncratic collection. These are the standard forms that outer certificates take across formal mathematics, proof architecture, and structured representation theory. The theorem stack therefore applies broadly, not only to the specific instantiation branches studied here.

The key point is negative as much as positive: *no outer certificate form built from the declared grammar of normal forms can escape the outer gate.* The grammar-cut maximality theorem is a completeness result for the class of normal-form outer certificates. Everything built from NF0–NF3 is accounted for.

The broader significance is this: this paper is not classifying a menu of admissibility witnesses. It is proving the *architecture of outer admissibility itself* — the forced structure that any sufficiently disciplined outer certificate world must have. The three levels are not three outcomes of one particular construction. They are three kinds of structural content that the logic of admissibility forces into three distinct bands.

12.2 Three instantiation branches

The architecture has three concrete instantiation branches:

- The *halting/semantic* branch (NEMS), which provides halting-anchored semantic summit certificates.
- The *infinity-compression* branch (IC), which provides compression carriers and native stage-D structure.
- The *continuation uniformization* branch (APS), which provides indexed middle-layer structure via recursion uniformization.

Each branch supplies a distinct route; the three-branch closure theorem `three_branch_corpus_highest_summit_at` together with Theorem 5.1 show these routes collapse to the same outer gate. The outer gate theorem is therefore not branch-local. It is a theorem about the three-branch architecture as a whole: distinct structural routes into admissibility collapse to the same gate.

12.3 Connection to the foundational laws

The outer admissibility results are not isolated. Each foundational law finds its reflection here: no free collapse means the gate requires a genuine pullback witness; burden relocates means the three arms all relocate into the same pullback structure; route architecture is forced means the second layer L_2 is exactly what admission forces above itself; and the residual condition's irreducibility (F_2) is the machine-checked form of "if the burden is invisible, suspect the map." The three-level architecture is the outer-certificate expression of a principle that was present from the foundations.

13 What lies beyond the summit

The architecture is complete at its solved level, but not globally exhaustive. Three frontiers are honestly open.

13.1 Global unconditional frontier

The existence of an absolute frontier S1 certificate for an arbitrary predicate–burden pair without hypotheses — $\mathrm{AbsRawS1}(P, B)$ for all P, B — remains open. This is the next genuine research frontier beyond the outer gate program [1]. It is not a missing piece of the current architecture; it is where the current architecture points next.

13.2 Broader outer grammar

If a principled grammar larger than the current NF0–NF3 normal-form grammar were ever chartered, the grammar-cut maximality result would need re-examination in that scope. The NF4 boundary finding is not a proof that no broader grammar is possible. It is a proof that the nearest specific candidate (NF4) did not force one now. A future investigation could, in principle, extend the grammar and re-examine the ceiling.

13.3 Residual condition universality

Whether the residual condition StageD could ever be brought inside the outer gate for all γ — i.e., whether $U_{\text{pullback}} \Rightarrow$ StageD universally — or whether the branch-specific obstruction (F_2) is genuinely permanent, is open. The residuality result gives "partial basis, irreducible residue." Establishing $U_{\text{pullback}} \Rightarrow$ StageD universally would be a genuine extension of the current theorem stack.

13.4 What these open frontiers mean

These three frontiers are not gaps in the current theorem stack. They are the next genuine research questions, now clearly located by the architecture. The architecture does not hide them; it exposes them precisely.

A solved architecture is valuable partly because it tells you exactly where it stops. The present theorem stack does not hide its frontier; it exposes it.

14 Machine verification and reproducibility

14.1 The Lean library

The Adequacy Architecture is formalized in a Lean 4 library [2] using Mathlib [3]. The underlying Lean library contains a large machine-checked corpus of named definitions, structures, and theorems spanning the foundational adequacy laws, the branch instantiations, the outer-gate program, and the layered consequences proved here. The principal outer-gate spine is tabulated in Appendix A. A complete named-theorem index appears in `THEOREM_INVENTORY.md` in the `adequacy-architecture-lean` repository tree.

14.2 The machine-checked status

All theorems stated in this paper are machine-checked in Lean 4 with 0 `sorry`. Toolchain: `leanprover/lean4:v4.29.0-rc6`, Mathlib `v4.29.0-rc6`.

To reproduce: run `lake build` from `adequacy-architecture-lean/`. The build succeeds with 0 `sorry`. The reproducibility script is `scripts/verify-lean-build.sh`.

14.3 The companion note

The companion note (*Technical Companion: The Solved Theorem Stack of the Adequacy Architecture*) gives the same theorems in a concise, theorem-dense format suitable for technically fluent readers who need exact Lean names and proof-sketch references without the motivating framing of this paper.

The two documents are designed to be read together. This paper provides altitude and conceptual orientation; the companion provides precision and artifact accountability.

Acknowledgements

All machine-checked proofs are in the `adequacy-architecture-lean` repository. A full theorem-name index and reproducibility instructions are included in that repository. See [1] for the complete program record and the mathematical kernel statement.

A Theorem table: outer gate stack

All modules are under `adequacy-architecture-lean/AdequacyArchitecture/`. All theorems machine-checked; 0 `sorry`.

Omitted row (S1-shape counterexample). See §6: the counterexample `not_s1LawfulFrontierRow_counterexample_pair` (`CertifiedFrontierRow.lean`) is load-bearing for the NF4 discussion but is omitted here so the table lists only collapse, maximality, layer, and residual arrows.

Result	Lean name	Module
Thm. 5.1	`program1FiniteGAdm_iff_program1AdmissibilityPullbackDisplayWitness`	`Program1MetaUnificationMU1.lean`
Thm. 5.2	`program1FiniteGAdm_implies_exists_absoluteFrontierRawS1`	`Program1MetaUnificationMU1.lean`
Thm. 6.1	`program1_outer_grammar_cut_maxA`	`Program1OuterGateGrammar.lean`
Thm. 6.2	`Program1OuterCertificateSound_program1OuterWinS1LawfulNonempty`	`Program1MaximalityMA1.lean`
Thm. 7.1	`program1_layer2PullbackDisplay_with_host_summitFE3_of_U_pullback`	`Program1LayeredStructureLS1.lean`
Thm. 7.1	`program1_layer2PullbackDisplay_with_host_summitFE3_of_program1FiniteGAdm`	`Program1LayeredStructureLS1.lean`
Thm. 8.1	`icSpec031_stageD_resolution_pack`	`ProgramFiniteAdmissibility.lean`
Prop. 8.2 F_1	`icSpec031_stageD_iff_conjunct_of_af1_native_carrier`	`ProgramFiniteAdmissibility.lean`
Prop. 8.2 F_3	`icSpec031_stageD_bypass_conjunct_of_af1_standard`	`ProgramFiniteAdmissibility.lean`
Prop. 8.2 F_4	`icSpec031_stageD_cascade_conjunct_of_af1_standard`	`ProgramFiniteAdmissibility.lean`

B Theorem table: foundational theorems

Selected foundational theorems that the outer gate program builds on. All machine-checked; 0 `sorry`.

Lean name	Module
`adequacy_interface_exists`	`Foundation/AdequacyInterface.lean`
`carrier_exists_of_adequacy_mode`	`Foundation/BurdenCarrierExistence.lean`
`internality_outsourcing_barrier`	`Foundation/InternalityOutsourcing.lean`
`no_free_adequate_collapse`	`Burden/NoFreeCollapse.lean`
`burden_not_erased_only_relocated`	`Burden/Relocation.lean`
`adequacy_preserving_drop_implies_relocation`	`Burden/Relocation.lean`
`universal_irreducible_adequacy_lawful`	`Lawful/MasterTheorem.lean`
`bottleneck_one_bundle`	`Lawful/BottleneckTheorems.lean`
`master_conditional_summit_ridge_at`	`Lawful/MasterConditionalSummit.lean`
`highestReachable_summit_at`	`Lawful/FinalConditionalSummit.lean`
`three_branch_corpus_highest_summit_at`	`Instances/ThreeBranchClosure.lean`
`standardRoleAwareRepresentationBundle`	`Instances/RoleAwareThreeBranchFrontier.lean`
`AbsoluteFrontierRawS1`, `HFrontier`	`AbsoluteFrontier/`
`every_map_has_fundamental_residual_package`	`Instances/WhatMapsForget.lean`
`lawfulAdequacyArchitecture_pullbackAlongCompare`	`Lawful/ComparePullback.lean`
`icSpec031_stageD_resolution_pack` (full four-conjunct decomp.)	`Instances/ICCompareRepresentationPullback.lean`
`CertifiedSummitRowRepresentation`	`Portability/RepresentationCalculusRL9.lean`
`certifiedSummitRowRepr_summitFE3_joint_upgrade_iff_row`	`Portability/RepresentationCalculusRL9.lean`
`summitReportOf_indexed_master_agreement_consequences`	`Instances/NEMSSemanticFaithfulTransport.lean`
`M1_from_certifiedFrontierRow`	`Portability/Spec034ProgramClosureLaw.lean`
`fe4_earned_absoluteFrontierRawS1_ladder_bundle`	`Portability/AbsoluteFrontierEarnedToy.lean`
`program1FiniteGAdm_icNems`	`Instances/ProgramFiniteAdmissibility.lean`

References

[1] N. Spivack. *Adequacy Architecture: Outer Gate, Layered Certificate Architecture, and Machine-Checked Synthesis.* Adequacy Architecture Project, `adequacy-architecture-lean` repository, 2026.

[2] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In A. Platzer and G. Sutcliffe, editors, *Automated Deduction — CADE 28*, volume 12699 of *LNCS*, pages 625–635. Springer, 2021.

[3] The Mathlib community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381. ACM, 2020. `https://github.com/leanprover-community/mathlib4`.